

# ActInf Textbook Group ~ Cohort 3 ~ Meeting 4

## (Chapter 2, part 1)

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**Duration:** 55:06 | **Uploaded:** Unknown **Transcript method:** auto\_caption

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all right hello it's February 15 2023 we're in cohort 3 of The Textbook group and we're in our first discussion on chapter two of the textbook so there's a lot to discuss chapter two goes hard before we go into the questions that people have raised does anyone want to just share a first recollection something that stuck with them or a friction or just a sentiment or a feeling that they had when they were reading even part of chapter two you can just unmute and go for if you want to share it's awesome that so many people added questions there's going to be a lot to discuss anyone just want to give a first thought what did they what did they hope chapter two would do was there any section or like quote or what stood out when people were on the low road to active inference okay let us skim in a few minutes the whole chapter and then we're gonna turn to the questions and just go based upon which questions people have voted as interesting but let's just take a quick uh flyby of the low road so chapter two is going to be coming on the low road so just to get a visual reminder of what we're talking about with the high road and the low road from figure 1.2 the low road is like the how showing how from Bayes theorem which is so simple and tautological and contentless we get a how that meets with a y which is this imperative for persistence and self-organization and they come together in the generative models used in active inference which we're going to get to in chapter four so the low road is going to talk about a lot of the how and some of the what I guess as well chapter three is going to come from The High Road perspective and chapter four is where we're going to come to the heart of active inference with the generative models thank you Maria we'll get to 2.3 and then feel free to address that section 2.2 is perception as inference and this section gives a little bit of an overview on how even before the Bayesian brain was described there was concepts of perception as inference with helmholtz and even further back as action Maria shared in live stream 43.0 and what active inference is doing is extending that inferential framework Beyond perception to also include action that is in the title it's active inference it's about inference on perception and action this is going to be

operationalized with modern statistics throwba is called a generative model we're going to come back through this so just moving rapidly probabilistic reasoning is described by this equation which is um it'll be awesome I think as a question asked like to have this in plain language what is being described or what is being done with basic question an example is introduced where a person is holding either a frog or an apple in their hand and then the object is going to either jump or not jump the person has some prior beliefs about a priori prior beliefs How likely frogs and apples are in the world and they have some beliefs about what is likely to do what action an observation comes in and then one is able to have a posterior belief or a posterior a posteriori of How likely they think the object is to be one thing or the other they've updated their prior beliefs through observation so like the Bayesian moment is this critical moment where a new observation comes in and that in the context of a likelihood model updates the prior into a posterior here's an exact Bayesian inference on that um scenario and for simple settings exact Bayesian inference is totally fine it turns out that for larger and more challenging statistical areas one has to use approximate Bayesian inference but what approximate Bayesian inference approximates is exact Bayesian inference so this is kind of like the core of what you could do if you had infinite computational power and you can do it for simple examples there's a discussion of probability distributions because there's multiple probability distributions as we're dealing with probabilistic variables people may be familiar with the gaussian or the so-called normal distribution or bell curve however there are other probability distributions that cover a different space on the x-axis that's called support and ultimately just have different shapes so sometimes you want to be reducing your uncertainty about a bell curve other times you might want to be reducing your uncertainty about other distribution families like waiting times there's a discussion of two kinds of surprise there's kind of plain surprise or surprisal which is just how surprising was that observation and it doesn't imply any updating of the generative model or belief it's just like taking a look at that observation alone how surprising was it we call that surprise surprisal and it's measured in information theoretic units a Nat is like a bit but a bit is like a zero or one coin flip a Nat is like a bit but it's like your coin has  $e$  sides 2.7 Etc sites so that's surprise just simply how surprising is an observation and then there's also Bayesian surprise Bayesian surprise also measured in Nats is about how much the observation updates the prior into the posterior so they're both measured in information theoretic units surprisable is describing um how surprising an observation is with no reference to the updating and the second notion of surprise is the Bayesian surprise which is how much beliefs update expectations are described

in box 2.2 although conversationally we might talk about like are you expecting it to rain tomorrow and expectation of a distribution is not a prediction about how it will be in the future necessarily expectation is the just arithmetic mean so it's just the center of gravity the center of mass of a probability distribution is its expectation having introduced a more General Bayesian and information theoretic perspective on just inference it's gonna return to biology so Maria asked what is um how does active inference relate with Biology and psychology was that the point of section 2.3 so does anybody want to give a thought like what was section 2.3 doing or how is optimality and Bayesian statistics related to biological inference either restating what they added or just adding your own angle on it Ali go for it uh I think section 2.3 uh serves as a kind of interlude between 2.2 namely perception as action and 2.4 action uh sorry perception as inference and 2.4 action as inference because uh before dealing with the concept of action as inference we need to be clear uh what kind of inference we're dealing with here namely uh it's not just uh any kind of uh inference that's for example been discussed in other various uh literatures and statistics or in other areas but we're dealing with some specific kinds of Bayesian inference uh which uh is uh one of its main characteristic is being optimal because uh to be optimal in this framework is a necessary condition for um for and for inference because uh without optimality we wouldn't have any um uh actionable um state of knowledge uh so um that's one of the main points of 2.3 uh because uh I believe the concept of optimality is a necessary requirement for any biological or even non-biological agent to act as a sentient agent in any kind of uh environment so that optimality is a kind of at least in this section it's been described as a kind of requirement for any kind of sentient Behavior thank you awesome great way to put it as like an interlude because biological or even the kinds of active systems that aren't necessarily biological we want to talk about inference on perception but also we want to talk about inference on action and part of the unifying capacity of active inference is that perception and action are going to be in service of a common imperative which is free energy it's a really important note that optimality does not imply that it's adaptive it doesn't mean the organism succeeds it doesn't mean it even works so it's really worth it to see what is meant by optimality is defined in relation to a cost function that's optimized that's the variational free energy which is going to be related to surprise which is why surprise was brought up as important earlier so it's also unpacked a lot more in the neuropsychiatry context where instead of saying well healthy or neurotypical people have optimal inference and then it's sub-optimal inference for those who are different in this or that way rather in the Precision neuropsychiatry setting we can talk about Bayesian inference being

optimal however with different generative models the outcomes of optimal inference are going to look different if somebody has an 100 belief that it rains every day then their optimal inference will lead to them believing that it rains every day so this doesn't mean Society accepts doesn't mean the business succeeds doesn't mean that the model works better than any other model it is just a statement about how Bayesian inference is going to unfold and as Ali pointed to this is helping us um move from a perceptual inference model into a subjective subject-specific generative model of perception and action which can be described as sentient in the sense used by Friston at all to mean adaptive and responsive to stimuli figure 2.2 is going to introduce a really important distinction between the generative process and the generative model so the generative model is the cognitive agent it's there whether you think of it as they are or they have a generative model that's like a realist angle or we're making a generative model for a cognitive agent that's the instrumentalist more map territory falling more on the map side so generative model is a Bayesian statistical model that is being used to describe a cognitive agent the generative process can be thought of as the niche that is the underlying process passing observations to the generative model the niche the generative process can be an active inference process itself but it doesn't have to be so the generative process could just be an if then it could be any kind of logic you can have any endogenous Dynamics however it passes observations through the Markov blanket as sent states to the generative model the GM section 2.4 the discussion to this point is common to all Bayesian brain theories we now introduce the simple but fundamental Advance offered by active inference alarm bells ringing it considers action as inference active inference is like plus pragmatic turn Bayesian inference perspective on action that gets unpacked a little bit more in the next two sections we clarify the single quantity that active inference agents minimize through perception and action is variational free energy so whereas a reward learning cognitive agents would be maximizing reward and active inference agent as modeled as the way we're going to be exploring in this textbook they have two ways to approach a single imperative the single imperative is variational free energy the two ways that variational free energy can be minimized is changing your mind and changing the world perception and learning and action so first just minimizing the discrepancy between the model and the world at a first approximation the common objective of perception and action can be formulated as a minimization of the discrepancy between the model and the world sometimes this is operationalized in terms of prediction error so here's a more classical prediction error minimization this is also reminiscent of like t-o-t-e Cycles looking for deviations and then acting to resolve deviations or discrepancies between

predictions of the generative model observations coming in from the generative process and then here are the two ways that discrepancies can be reduced perception change belief oh I didn't think it was raining but it's wet discrepancy alarm Bells surprise surprisal surprising observations so then what can be done well one can change belief so that upon the next cycle there is no discrepancy or there are situations where action can be taken so that future observations are in line with the unchanged predictions those two pathways are going to come back again and again inactive inference how much should we take the idea of cycle seriously from Jonathan presumably it's much more non-linear yeah that's an awesome question does anyone want to give a first thought on that there's there's a bunch of ways to explore this one is although sometimes we see it as a circle this is actually a very informal representation two representations or ways of thinking about it that are a bit closer to formality one we're going to come to in figure 4.3 and chapter 7 as well which is when you implement this computationally it's going to be like unrolled as a linear sequence so one answer is computationally you're going to unroll the ways that these variables update each other and they still May describe non-linear functions so we still can get a non-linear function being described at in various places even if it's unrolled and then the second formal that helps us escape this sort of like loops situation is there's going to be um not as much in this textbook but in a lot of the recent work on Bayesian mechanics equations are defined for these variables circles random variables as flows on those partitioned variables so in that setting you can draw the edges about how different variables flows influence each other but it's actually like an unfolding flow that just continuous across four different partitions that's the particular partition and so you don't need to draw any kind of like looping but the action perception Loop is commonly understood and it's also very deep topic uh sorry yeah I think uh for this question uh that figure from um path integrals strange kinds paper might also help to uh illuminate this uh concept of cycles and uh the relevance of this concept uh to different types of particles uh that have been described in this paper so yeah I just wanted to point it out to that diagram yes here's the flows on different states and then the diagram looking a little inverted but continue what would you like to say about it so basically uh in this paper they have defined four different types of particles the simplest one being inert particle which doesn't have any active States and as we increase uh in terms of um I mean complexity or uh increasingly more sentience Behavior we're dealing with active particles which have uh active States but uh as opposed to strange particles which do possess the disposition to being actually a sentient particles in case of active particles and conservative conservative particles there isn't any

hidden active States so that's basically uh encompasses uh the different self-organizing or even non-self organizing uh entities ranging from simple rocks to brains and organisms so I think that's a pretty good way to somehow uh a frame frame all of those behaviors within that Bayesian mechanical framework thank you ever I have a question of uh the figure 2.3 um it looks as if there's always discrepancy but I can also imagine that I have a perception of objects in the world say for example where there isn't any discrepancy between my prediction and my sensory observation but that doesn't is not shown by this picture or is it the discrepancy value could be zero and especially for in silico simple examples like where the generative model is the exact same structure as the generative process then the discrepancy can be zero I predicted a coin flip heads it came up heads no discrepancy I don't need to change my belief maybe I'll strengthen my belief and I don't need to take any further action so discrepancy value can be zero however in the context of fitting a generative that is a coarse grains or partial potentially noisy data coming in there's always going to be a non-zero discrepancy so that's why we're interested in relative discrepancy reduction but there's also contrived examples where the discrepancy could be down to zero but there are situations where whether the discrepancy is zero or low or already as minimal as it can be that there may be no perception change there may also be no action change or you could imagine observations coming in that induce large changes in belief High Bayesian surprise and are associated with the agent taking action thank you earlier in the case of the object jumping out of the hand or not there was equations showing exact Bayesian inference plug and chug just push the numbers through the Bayesian equations yet exact Bayesian inference supporting perception action is computationally intractable in most cases and they're going to describe the cases more it can just be understood coarsely as being hard to know the total State space of what could happen and it might require taking an integral or sums over functions that are very like large or intractable so it can be challenging to do exact Bayesian inference and so one of the key moves in active inference as most currently use it today is variational inference it will be unpacked in chapter four when we talk more about the generative models it suffices to say that performing variational Bayesian inference implies substituting the two intractable quantities posterior probability and log model evidence so that's like what we would want to know if we had full access to information infinite computational resources doable for simple problems intractable for higher dimensional State spaces and so on we're going to replace those two posterior probability is going to be replaced with an approximate posterior called  $Q$  and then log model evidence is going to be substituted functionally by a free energy calculation this is called energy-based learning

broadly it's been around for decades active inference does not invent the idea of using a free energy value on a Bayesian statistical learning or inference challenge what is different is inference on action but it is not something that was generated within the octave inference Community it's existed for decades variational free energy variational Bays is used broadly and every day octave inference uses inference on action and perception that's what integrates that kind of perceptual signal processing elements with the action selection control theoretic element and it uses the particular partition which is the partition of the cognitive agent from the niche to operationalize variational inference in this kind of agent-based perception cognition action Loop or just scenario so it's always important as we're like learning through and again multiple coats of paint but what is actually new and different in active inference what is describing or condensing or distilling work that is totally non-contentious and has existed for decades and is being used day to day evidence lower bound and energy-based learning people will find copious examples of so that's one deflationary take to use a fristianism on active inference is doing variational Bayesian inference isn't novel unified imperatives for perception and are also not necessarily novel but using variational inference on the particular partition so that there can be a unified imperative for perception cognition is when we actually get to active inference equation 2.5 describes variational free energy we're going to come back to that figure 2.4 gives a visual representation of how variational free energy bounds surprise so here's like what we would want to know here's how we can approximate or bound what we want to know figure 2.5 revisits that discrepancy yes ever at first yes I I have some difficulty with this founding uh can you explain to me yes this figure uh what is meant by PMG bounds what what about awesome so I'll just read what someone else has already added and then anyone else feel free to like raise your hand to add more so um there's something we can't actually calculate but we can calculate something approximately we know that whatever value we get for the approximate thing the True Value we're looking for is certainly lower than the thing we can calculate so we want to know if the organism's age you don't have any information about them but you do know the organism can never live to be over 10 years old so that's an upper bound on what you believe the age of the organism to be so surprisal so  $\ln$  is the natural log so natural log and you can undo and do natural logs with exponentiation not going to go into too much detail on that but it's just an operation like whether when you see something  $\ln$  and then parentheses you can um take it in and out we're going to talk more about logs later um so this is what we would really want to know how surprised should we be for a given observation because if we knew how truly surprised to be

for any given observation we would have access to that true underlying distribution that is controlling how surprised we should be like if you don't know the the range in temperature you don't know how surprised to be for different observations if you knew the underlying temperature distribution you would know exactly how surprised to be with different thermometer readings what energy-based learning does again this is not an active inference sm this is energy-based learning or evidence lower bound used broadly in Bayesian statistics it constructs an approximatable functional which is a function of a function an approximatable functional free energy variational free energy equation 2.5 that's a function of  $r$  approximation which is a function that's why  $f$  is a functional and data so instead of trying to find the absolute underlying surprise how surprised we should be by a data point which would entail knowing the true distribution we're going to construct this functional variational free energy that's a function of  $q$   $r$  approximation and  $Y$  the data and then this is a KL Divergence the double line what is on both of the sides the double line is what is being Divergence minimized we're trying to minimize the Divergence between our approximation and the true distribution conditioned on data this is guaranteed to be an upper bound it's always higher than the surprise and then by doing diversions minimization with KL Divergence it's possible to push that upper bound lower so it's like here's how much the bank account actually has okay I know it's less than a billion because that's there's not that many of this token in is it less than a thousand is it less than five so it's like as you push the upper bound down you're able this can be approximated well again whether your startup works or whether the organism lives or whether the model's adaptive or whether it's better than any other model anyone else could ever think of that's an it's not this question this is saying you can generate a derivative not in terms of like a rate of change but like a sort of corollary based around data and our approximation that's always higher than surprise so surprise was brought in early in the chapter as a key imperative to surprise minimize but then that's intractable so we have an approximatable tractable form and that's what energy-based learning is and it's not invented in active but what is a little novel and different is doing energy-based learning Bayesian variational inference we revisit in 2.5 figure and instead of each being discrepancy here which is kind of like an a prediction error minimization that's like the tote like is there discrepancy reduce the discrepancy well discrepancy between the predictions and the observations when you think discrepancy you're thinking like it's going to be like I predicted a seven I got a nine the difference was two that would be prediction error free energy is doing energy-based learning and so it's like a discrepancy because if the data coming in were exactly what was predicted this Divergence



would be zero and this term would basically kind of functionally do what you would think a prediction error would do in the situation where the prediction matches the observation but it turns out to also extend in a few other ways but the reason why they showed the discrepancy first in 2.3 and then you can change perception or change your action selection and then prediction observation come together calculate a discrepancy and then decisions are made cognitive decisions perception or like embodied decisions in that change the actual generative and then in 2.5 that gets built out a little bit beyond just prediction error type discrepancies like there's a four inch discrepancy between these two things into an information theoretic or a statistical quantity that's variational free energy  $f$  section 2.7 talks about expected free energy and planning as inference you can't talk about variational free energy on data points that haven't come in yet because their future observations and so expected free energy  $G$  is similar and it's explored in 2.8 it's similar in certain ways but importantly and like we return to this chapter after chapter so no worries how much of these equations look fluent to you right now one thing is these equations have natural language representations that people can also improve but especially like 2.5 for  $f$  you can read it and see the active inference ontology terms being used rather than variables and then in 2.6 it's um some part of it is written but like these are the kinds of augmentations that are always very welcome for people to make because they help us understand um expected free energy is like variational free energy but whereas variational free energy is real time and it's crunching our approximation with the incoming data expected free energy is calculated over policies which are sequences of action and it is going to be dealing with observations that haven't yet come in so this is where we get to policy selection in active inference by thinking about expected free energy which is like a prospective talking about the expected free energy for different possible policies which are sequences of actions there's some more discussion about special cases expected free energy when certain terms are basically zeroed out expected free energy becomes resembling other equations so this is like a common technique in math you have some equation and then like you can generalize it and then a special case of the general form is this and then another special case might be that so for example we'll come back next week or in the to what exactly the pieces are um representing but one can imagine that if pragmatic was irrelevant then  $G$  would reflect Information Gain of different policies if information gained were irrelevant like it were a totally discovered situation then  $G$  would reflect purely pragmatic so  $G$  is like a generalization it's a unified imperative under which certain special cases resemble variously information maximizing principles or utility maximizing principles Jonathan is there any

equivalent to discounting as in an RL context do you want to unpack what you mean there by discounting but then yes there's definitely some angles on that yeah sure so um in a reinforcement learning context um you know there's there's a clear aim which is to maximize the expected future returns um but generally that's done in a discounted way and so the agent is trying to maximize the discounted expected future returns such that rewards that it gets in the future are essentially worth less than rewards that it gets in the near term so I know that there isn't the same kind of idea of rewards as there is in in reinforcement learning and active inference context but I'm interested to know whether there is any sort of idea of discounting that states further into the future or somehow less important than states in the moment yes here's a um I'll I'll add that um uh question and then add in the um one Briston paper it turns out that yes it's possible to to get temporally discounted type Behavior so like a simple way to think about that would be if the um pragmatic Drive were urgent then you would see small pragmatic gain being approached at the loss of future gain short time Horizon models and Urgent desire Drive whereas under longer time Horizons longer time preferences then you see the kind of behavior that in utility driven decision-making is associated with what is called temporal discounting so the phenomena the cognitive behavioral phenomena exists and and can be recapitulated but it is not generated with the same way that it is in other places sure thank you 2.9 is the end of the low road variational free energy real time unfolding  $F$  of  $Q$  and  $Y$  variational free energy our approximate posterior  $q$   $q$  that's the one we control  $Y$  data just like a regression  $y$  equals  $MX$  plus  $b$   $q$  and  $Y$  discrepancy except it's a variational free energy it's real time expected free energy  $G$  of  $P_i$   $G$  is a function of policies and we're going to go into that more but it is regarding observations that haven't come in yet so it has a different structure but some rhyming with variational free energy what do they achieve together where did we get at the end of the low road we started with Bayes theorem generative model could be the the thing jumping out of the hand perception is inference that goes back to helmholtz but as livestream 43 unpacks this has been understood broadly across the world for thousands of years predictive coding helps operationalize perception as inference because it's like top-down predictions bottom-up sensory observation information that's why we don't perceive our blind spot to bring in action we have to talk about perception as inference because we do care about observations but we have to talk about observations that haven't actually happened yet and we have to talk about how our actions influence which observations are likely to occur perception and action as inference together make up the Bayesian brain which is like some people it means the brain does Bayesian computation for that's a realist take the instrumentalist take

would be we use Bayesian statistics to study the brain where exact Bays is intractable we can use variational base energy-based learning that's the end point of the low road into active inference summary active inference is a theory of how living artifacts underwrite their existence by minimizing surprise that's what we would truly want to know surprise if you knew how surprised to be you would have had to have known the actual underlying distribution so we usually can't do that so we have tractable proxy to surprise so just like we'd want to reduce our surprise because that would reflect having the most optimally trained but not over fit belief we want to minimize a proxy that we know is an upper bound so by minimizing the upper bounds we're getting closer and closer to what we really want to approximate the two ways to do that are perception and action they motivated this from a very first principles perspective and then extended what is often left only in the realm of perception into the domain of action more discussion is a construct of information Theory it's not necessarily equivalent to folk psychology and this comes up again and again we see it in the ontology for like half of the words these were words that are used broadly perception action belief surprise and sometimes they're perfectly overlapping with formal definitions other times there's a little bit of a Divergence and they close perception action minimize variational free energy and complementary waste that's real time active inference also minimizes expected prospectively this completes our journey along the low in chapter 3 we travel the high road which reaches the same conclusion on the basis of first principles and self-organization so we started from we have a Bayesian We Didn't Start From organisms need to exist but it turns out by connecting surprisal to self-organization or really just persistent repeated measurements we're going to get to the same place all right in our five remaining minutes just next week we're going to return to the to the questions but I hope that it's useful to also just walk through it because definitely it takes like multiple readings and interrogations and there's things that like you'd want to know in chapter two that are going to come up later or come up elsewhere in other papers there's just no single Road or path let alone one that would be appropriate for all Learners and their preferences so hopefully it was useful to like look over it and see it that way next week we're going to return a lot to the questions if anybody wants to do addition of questions it's always great you can jump into the answers and discourse as I see many do and add things you can upvote what is interesting you can enrich what people have written by just replacing where they use a plain text with something that you know could be tagged just like that so there's all kinds of ways that like wherever chapter two is sitting with you now either by generating new questions and discourse you can add notes on chapter um here and then

also in the math group which we haven't been Super Active on we've done some laying out of just the mathematical skeleton of chapter two as an example of how that could be done so but in our closing minutes just we can have any closing thoughts okay just all anyone can raise their hand but um so in relation to the temporal discounting and reinforcement learning Ali also wrote reinforcement learning models opposed to active are not applicable to perceptual tasks that do not involve Rewards or show significantly variable Behavior so there's there's a lot there in situations where pragmatic value isn't a key imperative even if that's just a sub-task like isocating is driven by Information Gain yes our eyes dwell on what might be seen as rewarding but it's Information Gain that is the imperative for the isocating and so where pragmatic value or reward or utility functions are either non-existent or or flat reinforcement has a hard time and that leads to all these contrived reinforcement bonuses and novelties oh maybe we'll pay them for this and we'll do that um and significantly and um learning that reward function is very difficult and learning that reward function especially in like a partially observable or dynamic or volatile environments ultimately you need to reward the learning anyway so then you end up coercing what we would just call epistemic Value into a reward learning framework so why not just have a generalization of reward learning where in fully observed situations pragmatic value can be unabashedly pursued but in partially observable environments or where the generative model is a simplified or coarse graining structure relative generative process why not just authentically pursue the epistemic value which is what you can do with  $g$  but can only be done in a contrived or an ad hoc way if you have reward as your unified imperative that's not to say reinforcement learning models are not effective that again that the agent can't play chess or that the startup wouldn't work it's just to be clear that the adequacy of a model in the world is an empirical question and especially in um a textbook the points are often being made in principle great well again thanks for all these questions I hope that we can um rejoin next week with some thoughts and ends resources and other questions added in there's just going to be a lot of fun ways to take these different questions and we'll come back to them so thank you everyone and if you want um in the following hour you can come to the Discord and we're just going to continue just talking and chilling about education so thank you and farewell